

```
import random

# --- Simulated stored credentials ---
CORRECT_USERNAME = "user123"
CORRECT_PASSWORD = "pass123"

def generate_otp():
    """Generate a random 6-digit OTP."""
    return str(random.randint(100000, 999999))

def login():
    print("=== Login (Step 1: Password) ===")
    username = input("Enter username: ").strip()
    password = input("Enter password: ").strip()

    if username != CORRECT_USERNAME or password != CORRECT_PASSWORD:
        print("Invalid username or password.")
        return

    print("\nPassword verified successfully!\n")

    # --- Step 2: OTP generation and verification ---
    otp = generate_otp()
    print("=== Two-Factor Authentication (Step 2: OTP) ===")
    print(f"[SYSTEM] Your OTP is: {otp}") # Simulates sending OTP via SMS/email

    entered_otp = input("Enter the 6-digit OTP sent to your device: ").strip()

    if entered_otp == otp:
        print("\nLogin Successful")
    else:
        print("\nLogin Failed: Incorrect OTP.")

if __name__ == "__main__":
    login()
```

=== Login (Step 1: Password) ===

Enter username: user123

Enter password: pass123

Password verified successfully!

=== Two-Factor Authentication (Step 2: OTP) ===

[SYSTEM] Your OTP is: 494921

Enter the 6-digit OTP sent to your device: 494921

Login Successful